INORGANIC - ISOMERIZATION **April 5, 1997**

1. Discuss the structure (A) shown below with respect to the bonding of the thiocyanate ligands.

2. Give a labelled figure for the results of the figure (B) below after it undergoes one "Bailar twist".

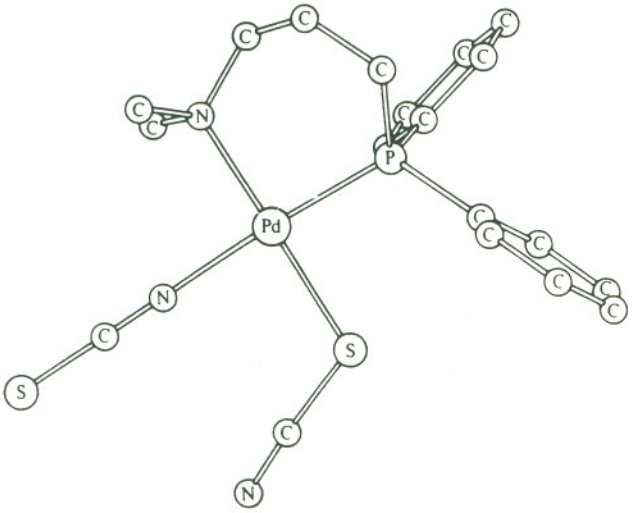
3. Sketch all of the isomers for $[CoF_3Cl_3]^{-3}$ if it is (a) octahedral and (b) trigonal prismatic

4. Explain how figures like (C) shown below are used in discussions of linkage isomerism.

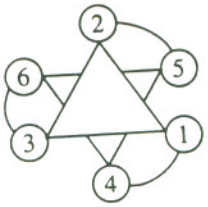
5. Give the complete names for all of the isomers of $[Co(NH_3)_3Cl_3]$.

6. How many isomers would you expect for Co(terpyridine)$Cl_3$ ? Give sketch(es).

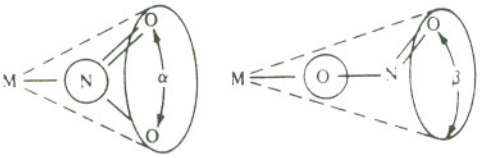
7. If you have a sample of optically pure $Cr(en)_3^{+3}$, how can you determine its absolute configuration?




A




B




C